GPS Tracker
Communication Protocol
(VT02F/VT03F/VT05F/VT08F)
V1.3

Content

1. Protocol Packet Format

Format	Length (Byte)	Description
Start Bit	2	0x78 0x78 (packet length: 1byte) 0x79 0x79 (packet length: 2bytes)
Packet Length	1(2)	Length = Protocol Number + Information Content + Information Serial Number + Error Check
Protocol Number	1	Transmission packet type (see the following diagram for details)
Information Content	N	The specific contents are determined by the protocol numbers corresponding to different applications.
Information Serial Number	2	The serial number of the first GPRS data (including status packet and data packet such as GPS, LBS) sent after booting is 1, and the serial number of data sent later at each time will be automatically added.
Error Check	2	Error check (From "Packet Length to Information Serial Number"),are values of CRC-ITU. CRC error occur when the received information is calculated, the receiver will ignore and discard the data packet. (See Appendix 3)
Stop Bit	2	Fixed value:0x0D0x0A

1.1 Protocol Number List

Protocol Number	Description
0x01	Login information
0x31	GPS Positioning data (UTC)
0x13	State information
0x21	String information
0x32	Alarm information
0x50	Multiple Bases Extension
0x80	Online command
0x33	Wifi information packet

2. Protocol Packet

2.1 Login Packet

2.1.1 Description

a) Login packet is the information packet connecting the terminal and platform. It can send terminal information to platform.
b) If a GPRS connection is established successfully, the terminal will send a first login message packet to the server within five seconds, if the terminal receives a data packet responded by the server, the connection is considered to be a normal connection; if not, the terminal will send login packet again.
c) If no packet returned by server within 5 seconds, then the response of login packet is timeout.
d) Terminal reboot automatically after 3 timeouts.

2.1.2 Login message packet

		Length	**Description**
Start Bit		2	0x78 0x78
Packet Length		1	Length = Protocol Number + Information Content + Information Serial Number + Error Check
Protocol Number		1	0x01
Information Content	Terminal ID	8	Example:IMEI number is 123456789123456 , terminal ID is:0x01 0x23 0x45 0x67 0x89 0x12 0x34 0x56
	Model Identification Code	2	Distinguish model of terminal by identification code.
	Timezone & Language	2	See the following chart for details of time zone language mark.
Information Serial Number		2	The serial number of the first GPRS data (including status packet and data packet such as GPS, LBS) sent after booting is ‘1’, and the serial number of data sent later at each time will be automatically added ‘1’ .
Error Check		2	Error check (From "Packet Length to Information Serial Number "),are values of CRC-ITU. CRC error occur when the received information is calculated, the receiver will ignore and discard the data packet. (See Appendix 3)
Stop Bit		2	Fixed value:0x0D 0x0A

Example:7878 11 01 0123456789123456 0000 3202 0001 D6FB 0D0A

2.1.3 Timezone & Language

<table>
<tr><td rowspan="12">One and a half bits
bit15 bit4</td><td>15</td><td rowspan="12" colspan="2">Timezone value expands 100times</td></tr>
<tr><td>14</td></tr>
<tr><td>13</td></tr>
<tr><td>12</td></tr>
<tr><td>11</td></tr>
<tr><td>10</td></tr>
<tr><td>9</td></tr>
<tr><td>8</td></tr>
<tr><td>7</td></tr>
<tr><td>6</td></tr>
<tr><td>5</td></tr>
<tr><td>4</td></tr>
<tr><td rowspan="4">Lower half bit4- bit0</td><td>3</td><td colspan="2">GMT(0:Eastern time, 1:Western time)</td></tr>
<tr><td>2</td><td colspan="2">No definition</td></tr>
<tr><td>1</td><td>Language Select Bit</td><td>Chinese</td></tr>
<tr><td>0</td><td>Language Select Bit</td><td>English</td></tr>
</table>

Bit3 0:Eastern time, 1:Western time

Example: Extended bit: 0x32 0x00 means GMT+8

Calculation method: 8*100=800 converts to HEX: 0x0320

Extended bit: 0x4D 0xD8 means GMT-12:45

Calculation method: 12.45*100=1245 converts to HEX: 0x04 0xDD

Here, to save 4 bytes, calculation result left shifted 4 bits and combined eastern time, western time and language bit.

2.1.4 Login packetresponse(server response)

	Length	Description
Start Bit	2	0x78 0x78
Packet Length	1	Length = Protocol Number + Information Content + Information Serial Number + Error Check
Protocol Number	1	0x01
Information Serial Number	2	Serial number of data sent later each time will be automatically added 1.

Error Check	2	Error check (From "Packet Length to Information Serial Number"),are values of CRC-ITU. CRC error occur when the received information is calculated, the receiver will ignore and discard the data packet. (See Appendix 3)
Stop Bit	2	Fixed value:0x0D 0x0A

Example:78 78 05 01 00 05 9F F8 0D 0A

2.2 Heartbeat Packet

2.2.1 Description

a) Heartbeat packet is a data packet to maintain the connection between the terminal and the server.
b) If a GPRS connection is established successfully, the terminal will send a first login message packet to the server and, within five seconds, if the terminal receives a data packet responded by the server, the connection is considered to be a normal connection; if not, the terminal will send login packet again.
c) If no packet returned by server within 5 seconds, then the response of heartbeat packet is timeout.
d) Terminal reboot automatically after 3 timeouts.

2.2.2 Heartbeat packet sent by terminal

Heartbeat		Length	Description
Start Bit		2	0x78 0x78
Packet Length		1	Length = Protocol Number + Information Content + Information Serial Number + Error Check
Protocol Number		1	0x13
Information Content	Terminal Information Content	1	See the following diagram for details
	Voltage Level	1	0x00:No Power (shutdown) 0x01 :Extremely Low Battery (not enough for calling or sending text messages, etc.) 0x02:Very Low Battery (Low Battery Alarm) 0x03:Low Battery (can be used normally) 0x04:Medium 0x05:High 0x06:Full
	GSM Signal Strength	1	0x00: no signal; 0x01: extremely weak signal; 0x02: weak signal; 0x03: good signal; 0x04: strong signal.
	Language/Extended Port Status	2	latter bit 0x01 Chinese 0x02 English

Serial Number	2	Serial number of data sent later each time will be automatically added 1.
Error Check	2	Error check (From "Packet Length to Information Serial Number "),are values of CRC-ITU. CRC error occur when the received information is calculated, the receiver will ignore and discard the data packet. (See Appendix 3)
Stop Bit	2	Fixed value: 0x0D 0x0A

Example:78 78 0A 13 40 04 04 00 01 00 0F DC EE 0D 0A

2.2.3 Terminal Information

One byte is consumed defining for various status information of the mobile phone.

Bit	Code Meaning	
Byte	Bit7	1:Oil and electribity disconnected
		0:Oil and electricity connected
	Bit6	1: GPS tracking is on
		0: GPS tracking is off
	Bit3~Bit5	Extended Bit
	Bit2	1: Charge on
		0: Charge off
	Bit1	1:ACC high
		0:ACC Low
	Bit0	1: Defense Activated
		0:Defense Deactived

2.2.4 Server responds the heartbeat packet

	Length (Byte)	Description
Start Bit	2	0x780x78
Packet Length	1	Length = Protocol Number + Information Content + Information Serial Number + Error Check
Protocol Number	1	0x13
Serial Number	2	Serial number of data sent later at each time will be automatically added 1.
Error Check	2	Error check (From "Packet Length to Information Serial Number "),are values of CRC-ITU. CRC error occur when the received information is calculated, the receiver will ignore and discard the data packet. (See Appendix 3)
Stop Bit	2	Fixed value: 0x0D0x0A

Example :78 78 05 13 01 00 E1 A0 0D 0A

2.3 GPS Location Packet

2.3.1 Description

a) Data packet used to transmit terminal location
b) Upload locating data based on rule after successfully connected and positioned.
c) Re-upload locating data after successfully connected.

2.3.2 Location packet sent by terminal

		Length	Description
Start Bit		2	0x78 0x78
Packet Length		1	Length = Protocol Number + Information Content + Information Serial Number + Error Check
Protocol Number		1	0x31(UTC)
Information Content	Date Time	6	Year(1byte) Month(1byte) Day(1byte) Hour(1byte) Min(1byte) Second(1byte) (converted to decimal) (Date Time) Timezone is UTC+0(Fixed)
	Quantity of GPS satellites	1	The first character is GPS information length. The second character is positioning satellite number (converted to a decimal)
	Latitude	4	Convert to a decimal and divide 1800000
	Longitude	4	Convert to a decimal and divide 1800000
	Speed	1	KM/H,Convert to a decimal
	Course, Status	2	Convert to binary number of 16 bits and calculate by bits (see the following diagram)
	MCC	2	Mobile Country Code(MCC) (converted to a decimal)
	MNC	2	Mobile Network Code(MNC)(converted to a decimal)
	LAC	2	Location Area Code (LAC) (converted to a decimal)
	Cell ID	4	Cell Tower ID(Cell ID)(converted to a decimal)
	ACC	1	ACC Status ACC low: 00, ACC high: 01(not available for 06)
	Data Upload Mode	1	GPS data upload mode : 0x00 Upload by time interval 0x01 Upload by distance interval 0x02 Inflection point upload 0x03 ACC status upload
	GPS Real-Time Re-upload	1	0x00 Real time upload 0x01 Re-upload(06 series are excluded)
Serial Number		2	Serial number of data sent later at each time will be automatically added 1

Error Check	2	Error check (From "Packet Length to Information Serial Number”),are values of CRC-ITU. CRC error occur when the received information is calculated, the receiver will ignore and discard the data packet. (See Appendix 3)
Stop Bit	2	Fixed value:0x0D 0x0A

Example:7878222211040702061DC7026C58E70C371AF30614B901CC00249000121700020000 2
2602A0D0A

2.3.3 Course & Status

Two bytes are consumed, defining the running direction of GPS. The value ranges from 0°to 360° measured clockwise from north of 0°.

Byte1	Bit7	0
	Bit6	0
	Bit5	0:GPS real-time 1:differential positioning
	Bit4	1:GPS having been positioning 0:not positioned
	Bit3	0:East Longitude 1:West Longitude
	Bit2	0:South Latitude 1:North Latitude
	Bit1	Course
	Bit0	
Byte2	Bit7	
	Bit6	
	Bit5	
	Bit4	
	Bit3	
	Bit2	
	Bit1	
	Bit0	

For example: the value is 0x15 0x4C, the corresponding binary is 00010101 01001100,

	Bit	Description
BYTE_1 Bit7	0	
BYTE_1 Bit6	0	
BYTE_1 Bit5	0	real time GPS
BYTE_1 Bit4	1	GPS has been positioned
BYTE_1 Bit3	0	East Longitude
BYTE_1 Bit2	1	North Latitude
BYTE_1 Bit1	0	Course 332°(0101001100 in Binary, or 332 in decimal)
BYTE_1 Bit0	1	
BYTE_2 Bit7	0	
BYTE_2 Bit6	1	
BYTE_2 Bit5	0	
BYTE_2 Bit4	0	

BYTE_2 Bit3	1
BYTE_2 Bit2	1
BYTE_2 Bit1	0
BYTE_2 Bit0	0

2.3.4 Server location packet response

Location packet server no response

2.4 LBS Multiple Bases Extension Packet

2.4.1 Description

For transmission of data packet when device is not located

2.4.2 Terminal sent LBS multiple bases extension packet

			Length	Description
Start Bit			2	0x78 0x78
Packet Length			1	Length = Protocol Number + Information Content + Information Serial Number+ Error Check
Protocol Number			1	0x50
Information Content	DATE (UTC)		6	Year(1byte)Month(1byte)Day(1byte)Hour(1byte)Min(1byte) Second(1byte)(converted to a decimal)(Date Time)
	TA		1	Timing Advance, Value= "Actual time of signal from Mobile Station to Location base -Time of signal from Mobile Station to Location base supposed the distance is 0
	MCC		2	Mobile Country Code(MCC) (converted to a decimal)
	MNC		1	Mobile Network Code(MNC)(converted to a decimal)
	Cell number		1	The cell number
	Cell 1	LAC	2	Location Area Code (LAC) (converted to a decimal)
		Cell ID	3	Cell Town ID(Cell ID) (converted to a decimal)
		RSSI	1	Signal level of community,range 0x00～0xFF, 0x00Weakest signal 0xFFStrongest signal
	Cell 2	LAC	2	Same as LAC
		Cell ID	3	Same as Cell ID
		RSSI	1	Same as RSSI
	Cell 3	LAC	2	Same as LAC
		Cell ID	3	Same as Cell ID
		RSSI	1	Same as RSSI
	Cell 4	LAC	2	Same as LAC
		Cell ID	3	Same as Cell ID

		RSSI	1	Same as RSSI
	Cell 5	LAC	2	Same as LAC
		Cell ID	3	Same as Cell ID
		RSSI	1	Same as RSSI
Reserved			1	
Reserved			1	
Reserved			1	
Serial Number			2	Serial number of data sent later at each time will be automatically added 1.
Error Check			2	Error check (From "Packet Length to Information Serial Number "),are values of CRC-ITU. CRC error occur when the received information is calculated, the receiver will ignoreand discard the data packet. (See Appendix 3)
Stop Bit			2	Fixed value: 0x0D0x0A

Example: 7878 31 24 170A1A100003 00 01B0 23 04 309C00AA403F 309C0035103A 309C00833037 309C005C202C 000000000000000000 02C8 FB72 0D0A

2.5 Alarm Packet

2.5.1 Description

a) Transmit alarm content defined by terminal.

b) Server response and parse longitude and latitude into address and re-upload to terminal after receiving the alarm content.

c) Terminal send address to pre-set SOS number of device.

2.5.2 Alarm packet sent by terminal

Alarm packet

		Length	Description
Start Bit		2	0x78 0x78
Packet Length		1	Length = Protocol Number + Information Content + Information Serial Number + Error Check
Protocol Number		1	0x32(UTC)
Information Content	Date Time	6	Year(1byte)Month(1byte)Day(1byte)Hour(1byte)Min(1byte) Second(1byte)(converted to a decimal)(Date Time)
	Quantity of GPS information satellites	1	The first character is GPS information length， The second character is positioning satellite number(converted to a decimal)
	Latitude	4	Convert to a decimal and divide 1800000

	Longitude	4	Convert to a decimal and divide 1800000
	Speed	1	Convert to a decimal
	Course, Status	2	Convert to binary number of 16 bits and calculate by bits (see the following diagram)(same as GPS packet, see GPS packet for details)
	LBS length	1	LBS length in total (LBS length +MCC +MNC +Cell ID)
	MCC	2	Mobile Country Code(MCC) (converted to a decimal)
	MNC	2	Mobile Network Code(MNC)(converted to a decimal)
	LAC	2	Location Area Code (LAC) (converted to a decimal)
	Cell ID	4	Cell Tower ID(Cell ID)(converted to a decimal)
	Terminal Information	1	See the following diagram
	Voltage Level	1	0x00:No Power (shutdown) 0x01:Extremely Low Battery (not enough for calling or sending text messages, etc.) 0x02:Very Low Battery (Low Battery Alarm) 0x03:Low Battery (can be used normally) 0x04:Medium 0x05:High 0x06:Full
	GSM Signal Strength	1	0x00: no signal; 0x01: extremely weak signal; 0x02: weak signal; 0x03: good signal; 0x04: strong signal.
	Alarm/Language	2	See the following diagram
Serial Number		2	Serial number of data sent later at each time will be automatically added 1.
Error Check		2	Error check (From "Packet Length to Information Serial Number "),are values of CRC-ITU. CRC error occur when the received information is calculated, the receiver will ignore and discard the data packet. (See Appendix 3)
Stop Bit		2	Fixed value: 0x0D0x0A

Example:78 78 25 26 0F 0C 1D 03 0B 26 C9 02 7A C8 18 0C 46 58 60 00 04 00 09 01 CC 00 28 7D 00 1F 71 80 04 04 13 02 00 0C 47 2A 0D 0A

2.5.3 Terminal Information

Bit	Code Meaning	
Byte	Bit7	1:Oil and electribity disconnected
		0:Oil and electricity connected
	Bit6	1: GPS tracking is on
		0: GPS tracking is off
	Bit3~Bit5	100:SOS

		011:Low Battery Alarm
		010:Power Cut Alarm
		001:Vibration Alarm
		000:Normal
	Bit2	1: Charging
		0: Not Charge
	Bit1	1:ACC high
		0:ACC Low
	Bit0	1:Defense Activated
		0:Defense Deactived

2.5.4 Alarm language

Byte 1	0x00:normal
	0x01:SOS
	0x02:Power cut alarm
	0x03: Vibration alarm
	0x04:Enter fence alarm
	0x05:Exit fence alarm
	0x06 Over speed alarm
	0x09 Moving alarm
	0x0A Enter GPS dead zone alarm
	0x0BExit GPS dead zone alarm
	0x0C Power on alarm
	0x0D GPS First fix notice
	0x0E Low battery alarm
	0x13 Remove alarm
Byte2	0x01Chinese
	0x02 English

2.5.5 Alarm packet response of server

	Length	Description
Start Bit	2	0x78 0x78
Packet Length	1	Length = Protocol Number + Information Content + Information Serial Number + Error Check
Protocol Number	1	0x26(UTC)
Information Serial Number	2	Serial number of data sent later at each time will be automatically added 1
Error Check	2	Error check (From "Packet Length to Information Serial Number"),are values of CRC-ITU. CRC error occur when the received information is calculated, the receiver will ignore and discard the data packet. (See Appendix 3)

Stop Bit	2	Fixed value:0x0D 0x0A

Example:78 78 05 26 00 1C 9D 86 0D 0A

2.6 Online command

2.6.1 Description

a) Use server online command to control terminal to execute task.
b) Terminal response results to server.

2.6.2 Online command sent by server

		Length	Description
Start Bit		2	0x78 0x78
Length of data bit		1	Length = Protocol Number + Information Content + Information Serial Number + Error Check
Protocol Number		1	0x80
Information Content	Length of Command	1	Server flag bit + command content length + language
	Server Flag Bit	4	Leave for server identification. Terminal receives the original data in Binary in response packet
	Command Content	M	Character string replied in ASCII coding. Command content is compatible with SMS command.
Information Serial Number		2	Serial number of data sent later at each time will be automatically added 1.
Error Check		2	Error check (From "Packet Length to Information Serial Number"),are values of CRC-ITU. CRC error occur when the received information is calculated, the receiver will ignore and discard the data packet. (See Appendix 3)
Stop Bit		2	Fixed value:0x0D 0x0A

Example :78 78 0E 80 08 00 00 00 00 73 6F 7323 00 01 6D 6A 0D 0A

2.6.3 Online command replied by terminal

	Length	Description
Start Bit	2	0x79 0x79
Length of data bit	2	Length = Protocol Number + Information Content + Information Serial Number + Error Check
Protocol Number	1	0x21

Information Content	Server Flag Bit	4	Leave for server identification. Terminal receives the original data in Binary in response packet
	Content Code	1	0x01 ASCⅡcode 0x02 UTF16-BE code.
	Content	M	Data needed to be sent(according to content code format)
Information Serial Number		2	Serial number of data sent later at each time will be automatically added 1.
Error Check		2	Error check (From "Packet Length" to Information Serial Number") , are values of CRC-ITU. CRC error occur when the received information is calculated, the receiver will ignore and discard the data packet. (See Appendix 3)
Stop Bit		2	Fixed value:0x0D 0x0A

Example:79 79 00 9D 21 00 00 00 00 01 42 61 74 74 65 72 79 3A 34 2E 31 36 56 2C 4E 4F 52 4D 41 4C 3B 20 47 50 52 53 3A 4C 69 6E 6B 20 55 70 3B 20 47 53 4D 20 53 69 67 6E 61 6C 20 4C 65 76 65 6C 3A 53 74 72 6F 6E 67 3B 20 47 50 53 3A 53 65 61 72 63 68 69 6E 67 20 73 61 74 65 6C 6C 69 74 65 2C 20 53 56 53 20 55 73 65 64 20 69 6E 20 66 69 78 3A 30 28 30 29 2C 20 47 50 53 20 53 69 67 6E 61 6C 20 4C 65 76 65 6C 3A 3B 20 41 43 43 3A 4F 46 46 3B 20 44 65 66 65 6E 73 65 3A 4F 46 46 00 2E 26 DF 0D 0A

2.7 Wifi Packet

2.7.1 Description

For transmission of data packet when device is not located.

2.7.2 Wifi packet sent by terminal

Wifi packet

		Length (Byte)	Description
Start Bit		2	0x78 0x78
Length of data bit		1	Length = Protocol Number + Information Content + Information Serial Number + Error Check
Protocol Number		1	0x33
Information Content	Date Time	6	Year(1byte)Month(1byte)Day(1byte)Hour(1byte)Min(1byte) Second(1byte)(converted to decimal)(Date Time) Timezone is UTC+0(Fixed)
	MCC	2	Mobile Country Code(MCC) (converted to a decimal)
	MNC	2	Mobile Network Code(MNC)(converted to a decimal)
	LAC	2	Location Area Code (LAC) (converted to a decimal)
	CI	4	Cell Tower ID(Cell ID) (converted to a decimal)
	RSSI	1	Signal level of community,range 0x00～0xFF, 0x00Weakest signal` 0xFFStrongest signal
	NLAC1	2	Same as LAC
	NCI1	4	Same as CI
	NRSSI1	1	Same as RSSI
	...	...	...

	NLAC5	2	Same as LAC
	NCI5	4	Same as CI
	NRSSI5	1	Same as RSSI
	TA	1	Timing Advance, Value= "Actual time of signal from Mobile Station to Location base -Time of signal from Mobile Station to Location base supposed the distance is 0
	WiFi number	1	WIFI number, 0 means no wifi detected
	WIFI MAC1	6	Wifi's MAC
	WIFI Strength1	1	Wifi signal strength
	...	...	...
	WIFI MAC5	6	
	WIFI strength5	1	
Information Serial Number		2	Serial number of data sent later at each time will be automatically added 1.
Error Check		2	Error check (From "Packet Length" to Information Serial Number") , are values of CRC-ITU. CRC error occur when the received information is calculated, the receiver will ignore and discard the data packet. (See Appendix 3)
Stop Bit		2	Fixed value:0x0D 0x0A

2.7.3 Server reply

No need to reply.

2.8 Information Transmission Packet

2.8.1 Description

Terminal transmits all types of non-position data.

2.8.2 Information transmission packet sent by terminal

Information transmission		Length	Description
Start Bit		2	0x79 0x79
Packet Length		2	Length = Protocol Number + Information Content + Information Serial Number + Error Check
Protocol Number		1	0x94
Information Content	Information Type (Sub-protocol Number)	1	00:External power voltage 0A:ICCID information
	Data Content	N	Different information type results in different transmission content. See the following for details.
Serial Number		2	Serial number of data sent later each time will be automatically added 1.

Error Check	2	Error check (From "Packet Length to Information Serial Number "),are values of CRC-ITU. CRC error occur when the received information is calculated, the receiver will ignore and discard the data packet. (See Appendix 3)
Stop Bit	2	Fixed value: 0x0D 0x0A

Example:

79790020940A08698141011151230460042429802256898604041018C0002255000571310D0A

Transmitted information content
When type is 00, the bit transmit external battery.
This bit is two-digit hexadecimal value. Hexadecimal value converted to decimal value and divide 100.
Example：0X04,0XC5, 04C5 converted to decimal is 1221， then divide 100 is 12.21, which means external voltage is 12.21V

When the information type is 0A, It means transmit ICCID, which is hexadecimal,

IMEI	8 bytes	0x08 0x69 0x81 0x41 0x01 0x11 0x51 0x23 means IMEI is 0869 814101115123
IMSI	8 bytes	0x04 0x60 0x04 0x24 0x29 0x80 0x22 0x56 means IMSI is 0460042429802256
ICCID	10 bytes	0x89 0x86 0x04 0x04 0x10 0x18 0xC0 0x00 0x22 0x55 means the ICCID is 898604041018 C0002255

2.8.3 Server responds packet

	Length (Byte)	Description
Start Bit	2	0x79 0x79
Packet Length	2	Length = Protocol Number + Information Content + Information Serial Number + Error Check
Protocol Number	1	0x94
Information Type(Sub-protocol Number)	1	The same with the device sent package sub-protocol number.
Serial Number	2	Serial number of data sent later at each time will be automatically added 1.
Error Check	2	Error check (From "Packet Length to Information Serial Number "),are values of CRC-ITU. CRC error occur when the received information is calculated, the receiver will ignore and discard the data packet. (See Appendix 3)
Stop Bit	2	Fixed value: 0x0D0x0A

3 Appendix

code fragment of the CRC-ITU lookup table algorithm implemented based on C language

```
staticconstU16crctab16[]=
{
0X0000,0X1189,0X2312,0X329B,0X4624,0X57AD,0X6536,0X74BF,
0X8C48,0X9DC1,0XAF5A,0XBED3,0XCA6C,0XDBE5,0XE97E,0XF8F7,
0X1081,0X0108,0X3393,0X221A,0X56A5,0X472C,0X75B7,0X643E,
0X9CC9,0X8D40,0XBFDB,0XAE52,0XDAED,0XCB64,0XF9FF,0XE876,
```

```
0X2102,0X308B,0X0210,0X1399,0X6726,0X76AF,0X4434,0X55BD,
0XAD4A,0XBCC3,0X8E58,0X9FD1,0XEB6E,0XFAE7,0XC87C,0XD9F5,
0X3183,0X200A,0X1291,0X0318,0X77A7,0X662E,0X54B5,0X453C,
0XBDCB,0XAC42,0X9ED9,0X8F50,0XFBEF,0XEA66,0XD8FD,0XC974,
0X4204,0X538D,0X6116,0X709F,0X0420,0X15A9,0X2732,0X36BB,
0XCE4C,0XDFC5,0XED5E,0XFCD7,0X8868,0X99E1,0XAB7A,0XBAF3,
0X5285,0X430C,0X7197,0X601E,0X14A1,0X0528,0X37B3,0X263A,
0XDECD,0XCF44,0XFDDF,0XEC56,0X98E9,0X8960,0XBBFB,0XAA72,
0X6306,0X728F,0X4014,0X519D,0X2522,0X34AB,0X0630,0X17B9,
0XEF4E,0XFEC7,0XCC5C,0XDDD5,0XA96A,0XB8E3,0X8A78,0X9BF1,
0X7387,0X620E,0X5095,0X411C,0X35A3,0X242A,0X16B1,0X0738,
0XFFCF,0XEE46,0XDCDD,0XCD54,0XB9EB,0XA862,0X9AF9,0X8B70,
0X8408,0X9581,0XA71A,0XB693,0XC22C,0XD3A5,0XE13E,0XF0B7,
0X0840,0X19C9,0X2B52,0X3ADB,0X4E64,0X5FED,0X6D76,0X7CFF,
0X9489,0X8500,0XB79B,0XA612,0XD2AD,0XC324,0XF1BF,0XE036,
0X18C1,0X0948,0X3BD3,0X2A5A,0X5EE5,0X4F6C,0X7DF7,0X6C7E,
0XA50A,0XB483,0X8618,0X9791,0XE32E,0XF2A7,0XC03C,0XD1B5,
0X2942,0X38CB,0X0A50,0X1BD9,0X6F66,0X7EEF,0X4C74,0X5DFD,
0XB58B,0XA402,0X9699,0X8710,0XF3AF,0XE226,0XD0BD,0XC134,
0X39C3,0X284A,0X1AD1,0X0B58,0X7FE7,0X6E6E,0X5CF5,0X4D7C,
0XC60C,0XD785,0XE51E,0XF497,0X8028,0X91A1,0XA33A,0XB2B3,
0X4A44,0X5BCD,0X6956,0X78DF,0X0C60,0X1DE9,0X2F72,0X3EFB,
0XD68D,0XC704,0XF59F,0XE416,0X90A9,0X8120,0XB3BB,0XA232,
0X5AC5,0X4B4C,0X79D7,0X685E,0X1CE1,0X0D68,0X3FF3,0X2E7A,
0XE70E,0XF687,0XC41C,0XD595,0XA12A,0XB0A3,0X8238,0X93B1,
0X6B46,0X7ACF,0X4854,0X59DD,0X2D62,0X3CEB,0X0E70,0X1FF9,
0XF78F,0XE606,0XD49D,0XC514,0XB1AB,0XA022,0X92B9,0X8330,
0X7BC7,0X6A4E,0X58D5,0X495C,0X3DE3,0X2C6A,0X1EF1,0X0F78,
};
//calculate the 16-bit CRC of data with predetermined length.
U16GetCrc16(constU8*pData,intnLength)
{
U16fcs=0xffff;//initialization
while(nLength>0){
fcs=(fcs>>8)^crctab16[(fcs^*pData)&0xff];
nLength--;
pData++;
}
return~fcs;//negated
}
```